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2.16 Exocytosis and Endocytosis

FlexBooks 2.0 > CK-12 Biology For High School > Exocytosis and Endocytosis

Written by: Douglas Wilkin, Ph.D. | Jean Brainard, Ph.D.

Fact-checked by: The CK-12 Editorial Team

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What you will learn

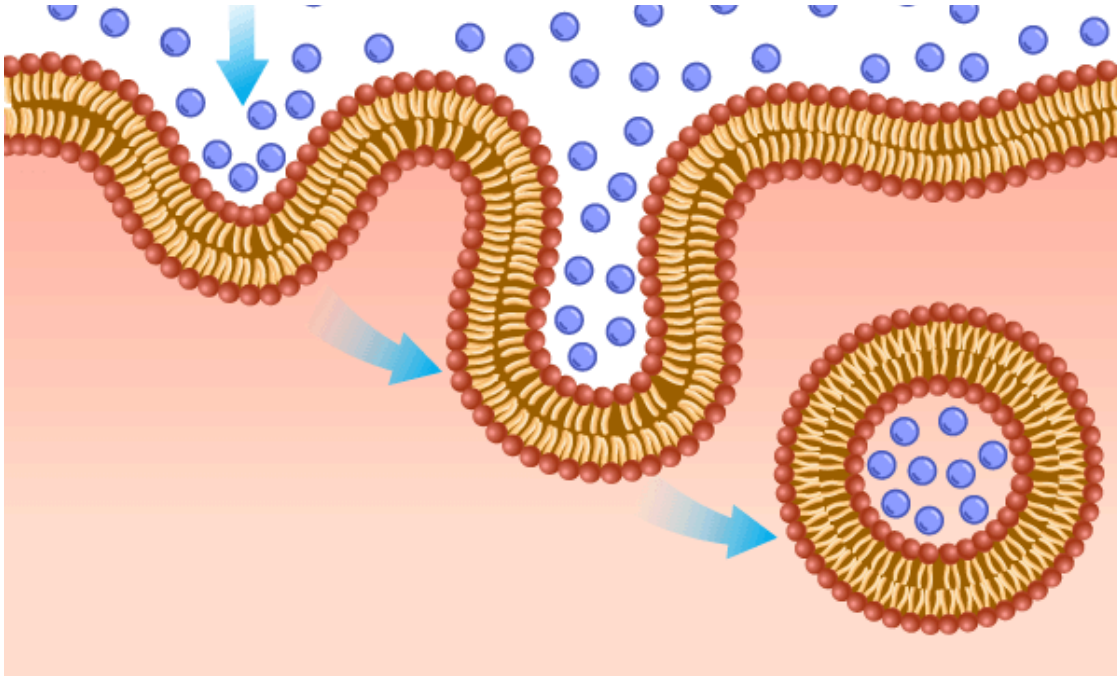
- How **vesicles** in the **cytoplasm** move large particles across the **cell membrane**
- What happens in **endocytosis** and exocytosis

Assign to class



Class Analytics





[Figure 1]

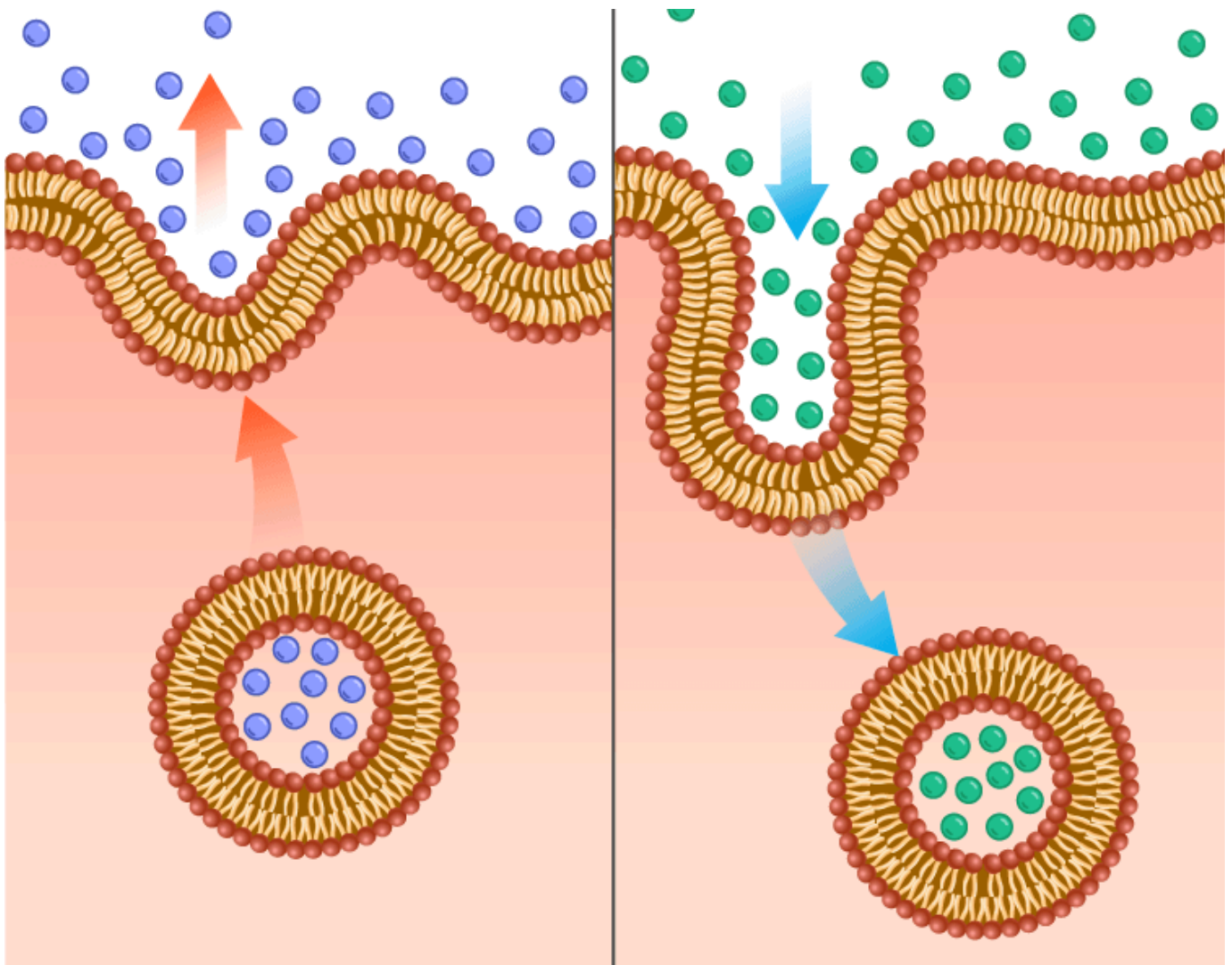
What does a cell "eat"?

Is it possible for objects larger than a small molecule to be engulfed by a **cell**? Of course, it is. For example, cells of the **immune system** consistently destroy **pathogens** by essentially "eating" them.

Vesicle Transport

Some molecules or particles are just too large to pass through the plasma membrane or to move through a **transport protein**. So cells use two other **active transport** processes to move these macromolecules (large molecules) into or out of the cell. Vesicles or other bodies in the cytoplasm move macromolecules or large particles across the plasma membrane. There are two types of **vesicle transport**, endocytosis and **exocytosis** (illustrated in the **Figure below**). Both processes are **active transport** processes, requiring **energy**.





[Figure 2]

Illustration of the two types of vesicle transport, exocytosis and endocytosis.

Endocytosis and Exocytosis

Endocytosis is the process of capturing a substance or particle from outside the cell by engulfing it with the cell membrane. The membrane folds over the substance and it becomes completely enclosed by the membrane. At this point a membrane-bound sac, or vesicle, pinches off and moves the substance into the **cytosol**. There are two main kinds of endocytosis:

- **Phagocytosis**, or *cellular eating*, occurs when the dissolved materials enter the cell. The plasma membrane engulfs the solid material, forming a phagocytic vesicle.
- **Pinocytosis**, or *cellular drinking*, occurs when the plasma membrane folds inward to form a channel allowing dissolved substances to enter the cell, as shown in the **Figure below**. When the channel is closed, the liquid is encircled within a pinocytic vesicle.



[Figure 3]

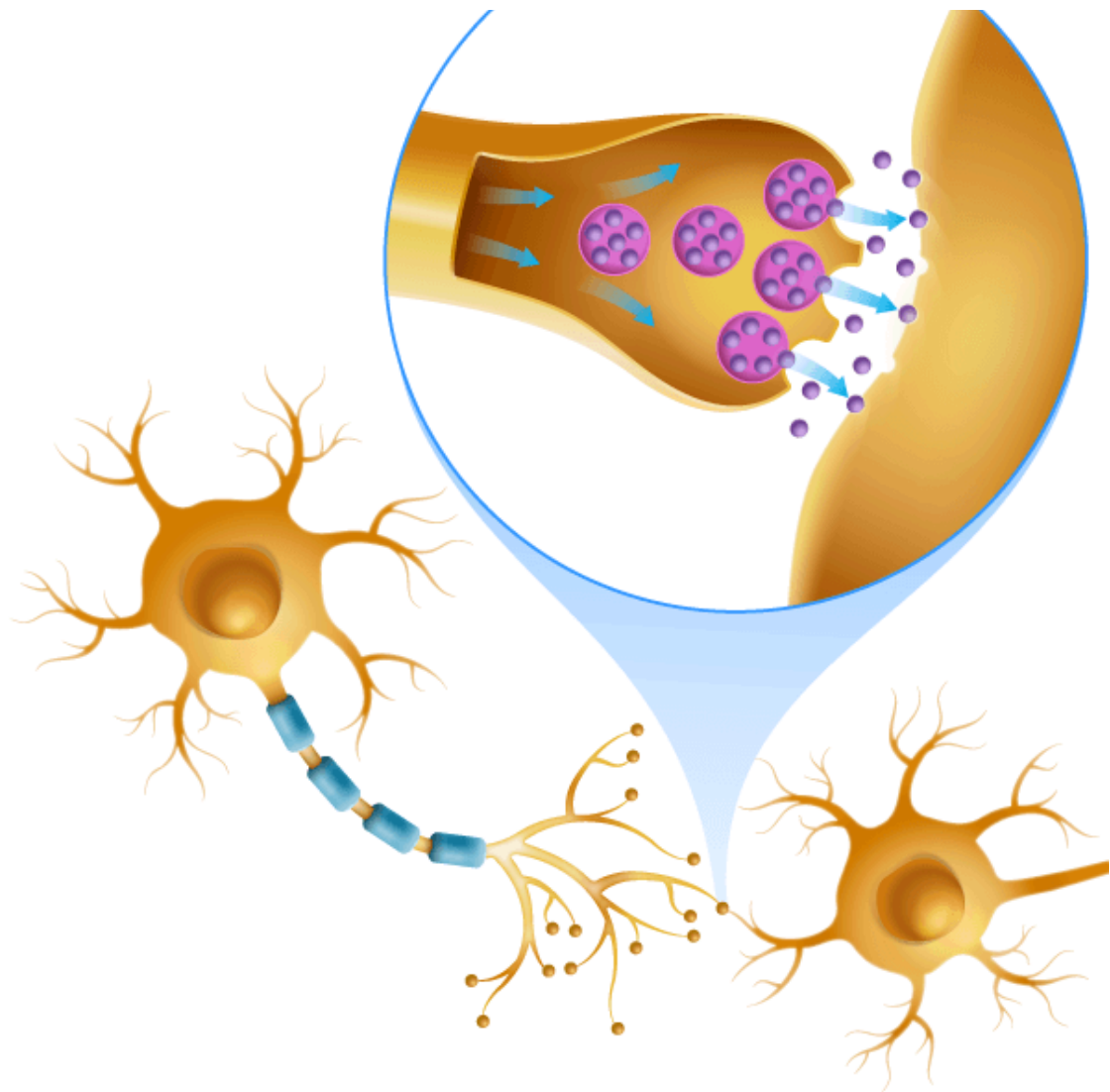
Transmission electron microscope image of brain tissue that shows pinocytotic vesicles. Pinocytosis is a type of endocytosis.

DID YOU KNOW?

Vesicles play an important role in capturing pathogens or other unknown substances that may endanger the **organism**. When pathogens are identified by the immune system, they are engulfed by immune cells to be destroyed.

Exocytosis describes the process of vesicles fusing with the plasma membrane and releasing their contents to the outside of the cell, as shown in the **Figure below**. Exocytosis occurs when a cell produces substances for export, such as a **protein**, or when the cell is getting rid of a waste product or a toxin. Newly made **membrane proteins** and membrane **lipids** are moved on top of the plasma membrane by exocytosis.





[Figure 4]

Illustration of an axon releasing a neurotransmitter by exocytosis.

DID YOU KNOW?

Exocytosis is a primary process involved in the cell membrane repair. When **transport vesicles** fuse with the cell membrane during exocytosis, they replenish the cell membrane's **phospholipid bilayer**.





membranes against a concentration gradient.

- Endocytosis is the process of capturing a substance or particle from outside the cell by engulfing it with the cell membrane and bringing it into the cell.
- Exocytosis describes the process of vesicles fusing with the plasma membrane and releasing their contents to the outside of the cell.
- Both endocytosis and exocytosis are active transport processes.

Review

1. What is the difference between endocytosis and exocytosis?
2. Why is pinocytosis a form of endocytosis?
3. Are vesicles involved in passive transport? Explain.



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